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The Additive Bachelier Model

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1 Introduction

In 1900, Louis Bachelier presented his doctoral thesis, entitled *Théorie de la spéculation* [1], under the supervision of the French mathematician Henri Poincaré. In this groundbreaking work, Bachelier developed the first mathematical model of Brownian motion and used it to derive a closed-form expression for the price of European options. This pioneering approach laid the foundation for the subsequent development of quantitative finance and marked the earliest application of probability theory to the study and analysis of financial markets. However, Bachelier's work received little recognition at the time and only gained widespread acceptance in the late 1960s, when financial mathematics began to emerge as a formal discipline.

Although the phenomenon of Brownian motion was first observed by the Scottish biologist Robert Brown in the early 19th century, a rigorous mathematical formulation did not emerge until several decades later, largely due to the work of Kiyosi Itô [7]. The lack of mathematical rigor in Bachelier's thesis is often cited as a reason why his ideas failed to capture the attention of both academia and the financial community for many years.

In contrast to Bachelier's largely overlooked work, the papers by Fischer Black, Myron Scholes [6], and Robert Merton [8] in the 1970s achieved immediate recognition, rapidly establishing the Black-Scholes model as the *de facto* standard for pricing European options. Their approach provided a rigorous mathematical framework for deriving a closed-form solution that, despite its academic novelty, closely mirrored the practical formulas already in use among traders at the time. This mathematical foundation helped bridge the gap between theory and practice, accelerating the adoption of quantitative methods in finance.

The original Black-Scholes model was subsequently extended by Black [2] to accommodate futures and commodity contracts, while Merton applied similar methods to the pricing of credit risk [9], further broadening the model's impact and solidifying its place at the core of modern financial theory.

The fundamental distinction between the two models, or rather families of models, lies in their core assumptions regarding the dynamics of the underlying asset. Let F_t denote the stochastic process describing the evolution of the asset's price over time. In the language of stochastic calculus, we can express this difference as follows:

$$\textbf{Bachelier: } dF_t = \sigma dW_t$$

$$\textbf{Black: } dF_t = \sigma F_t dW_t$$

where W_t is a Brownian motion under a proper probability measure. In practice, the former models the underlying as an *Arithmetic Brownian Motion (ABM)*, while the latter as a *Geometric Brownian Motion (GBM)*.

$$\text{Arithmetic Brownian Motion: } F_t = F_s + \sigma(W_t - W_s)$$

$$\text{Geometric Brownian Motion: } F_t = F_s e^{-\frac{1}{2}\sigma^2(t-s) + \sigma(W_t - W_s)}$$

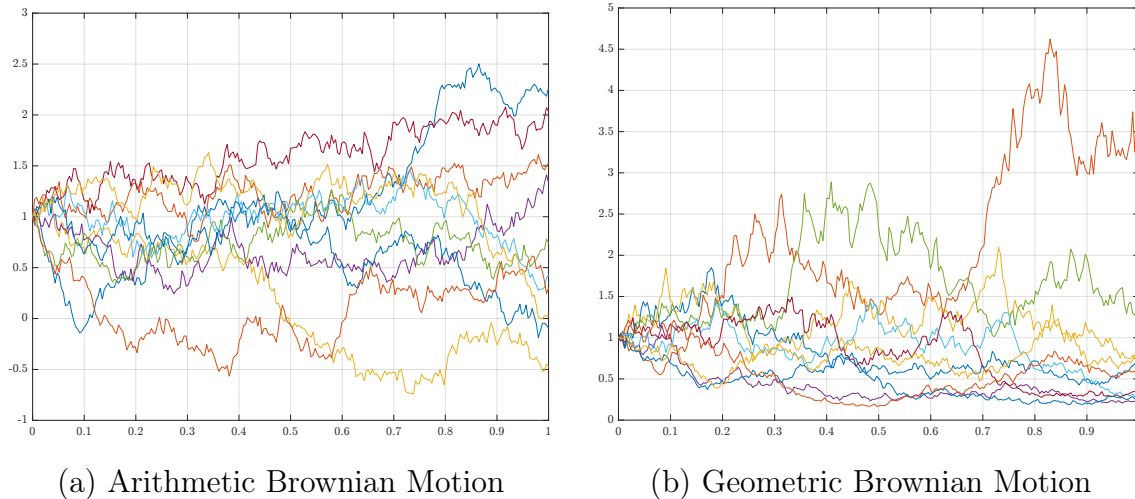


Figure 1: Monte Carlo simulation of the paths of the two processes.

As can be seen by the picture above, the two stochastic processes have a fundamental difference, with important financial implications. While the former trajectories can be either positive or negative, the Geometric Brownian Motion paths are strictly positive $\mathbb{P}$ -almost surely.

Positivity is often considered a desirable property when modeling stock prices; negative prices are typically deemed nonsensical. However, recent events have demonstrated that, in the context of commodity futures, prices can indeed become negative.

On April 20, 2020, the West Texas Intermediate (WTI) crude oil futures contract for May delivery experienced an unprecedented collapse. The price plummeted from a previous close of \$17.73 to an all-time low of $-\$37.63$ per barrel, marking the first instance in history where oil futures traded at negative prices [5] [14].

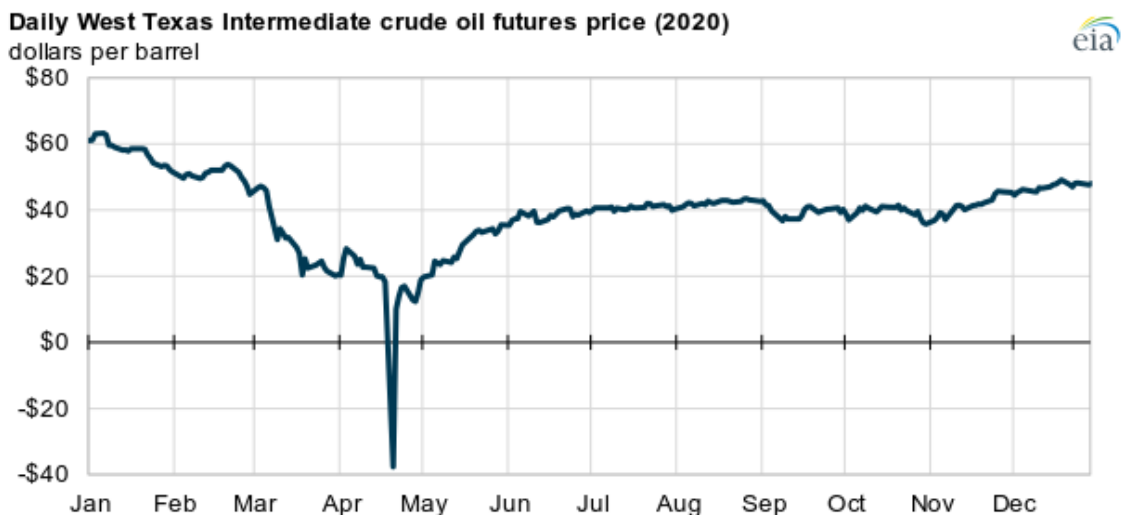


Figure 2: Source: U.S. Energy Information Administration, New York Mercantile Exchange (NYMEX)

This sudden falls was caused by a confluence of factors, including a shock to

global demand caused by the COVID-19 pandemic, an oversupply of oil, and limited storage capacity at the delivery point of Cushing, Oklahoma [3].

Other than the economical and political consequences, this event offers interesting take to rethink our conceptions about positivity of assets prices. In fact, all *Black-like* models used in derivative pricing on commodity contracts do not account for such possibility. Even weeks before oil futures went negative, the industry started evaluating such scenario and what actions could have been taken to ensure the functionality of markets. On April 8th, the Chicago Mercantile Exchange (CME), parent company of the NYMEX, sent a letter to all Clearing Member Firms, Chief Financial Officers and Back Office Managers with the title "*CME Clearing Plan to Address the Potential of a Negative Underlying in Certain Energy Options Contracts*" [4].

In the first chapter, regarding WTI futures, the letter states:

If WTI Crude Oil futures prices settle, in any month, to a price between \$8.00/bbl and \$11.00/bbl, CME Clearing may switch its pricing and margining options models from the existing models to the Bachelier model, currently utilized in numerous spread options products where negative underlying prices and strike levels are a regular occurrence. If any WTI Crude Oil futures prices settle, in any month, to a level below \$8.00/bbl, CME Clearing will move to the Bachelier model for all WTI Crude oil options contracts as well as all related crude oil options contracts effective the following trade date.

Oddly enough, after more than a century, the Bachelier seems to have found a much deserved importance in quantitative finance.

In this work, we consider European options data on WTI on June 2nd 2020, calibrating two different models: the Additive Bachelier model and the Additive Logistic model. The former is clearly an extension of Bachelier's original work. Our analysis is based on call and put options with maturities

Contract	Expiry
JUN20	2020/05/14
SEP20	2020/08/17
DEC20	2020/11/17
MAR21	2021/02/17
JUN21	2021/05/17
SEP21	2021/08/17
DEC21	2021/11/16
JUN22	2022/05/17
DEC22	2022/11/16

Table 1: Settlement dates for selected crude oil futures contracts. Since our analysis begins on June 2nd, the first batch of options expiring on May 14th is not actually taken into consideration

Before fitting the models—whose mathematical details will be presented in the following chapters—it is necessary to address the crucial step of recovering the discount factor. This is particularly relevant in commodity markets such as oil, where it is not immediately clear what kind of discounting rate market participants are actually using. To tackle this, we adopt a dedicated bootstrapping procedure, described in the next section.

Once the models are calibrated, we implement a Monte Carlo simulation framework to generate the underlying price paths. This allows us to price an exotic digital option whose payoff depends on an *Asian-like* average of the underlying. In the final section, we propose a quantitative hedging strategy for the contract, which demonstrates notable effectiveness even in the face of severe market dislocation.

2 Bootstrap

We were asked to calibrate forward prices and discount factors under the assumption that zero rates are a deterministic function of time. To achieve this, we employed the bootstrapping methodology outlined in [10].

We began by analyzing traded plain vanilla options—the most liquid instruments in the market—across various maturities. Specifically, we considered only those strikes K for which both a call and a put option were simultaneously traded. From these, we constructed a portfolio of *synthetic forwards* $\mathcal{G}$, defined as the composition of a long call and a short put position with the same strike.

According to the no-arbitrage principle, the prices of these synthetic forwards must satisfy *put-call parity*. If this condition is violated, one could exploit it via arbitrage strategies such as the *box spread*—a scenario sometimes observed in commodity markets.

Furthermore, the no-arbitrage framework implies that the forward price should be independent of the strike K . Empirically, we observed that the ratio $\mathcal{G}(K)/B(t_0, T)$ behaves approximately as a decreasing linear function of K . Based on this observation, we estimated the discount factor $B(t_0, T)$ as the *slope* of a linear regression:

$$\mathcal{G}_i = -\bar{B}(t_0, T)K_i + \bar{B}(t_0, T)F + \epsilon_i \quad \text{for } i = 1, \dots, N,$$

where $\mathcal{G}_i$ and K_i are the observed synthetic forward prices and strikes, respectively, and F is the forward price.

The estimator $\bar{B}(t_0, T)$ is then given by:

$$\bar{B}(t_0, T) = -\frac{\sum_{i=1}^N (K_i - \hat{K})(\mathcal{G}_i - \hat{\mathcal{G}})}{\sum_{i=1}^N (K_i - \hat{K})^2}$$

where

$$\hat{\mathcal{G}} := \frac{1}{N} \sum_{i=1}^N \mathcal{G}_i, \quad \hat{K} := \frac{1}{N} \sum_{i=1}^N K_i.$$

We then retrieved the zero rates using the relation:

$$\text{zRate} = -\frac{\log(B)}{\delta(t_0, T)}$$

where the year fraction $\delta(t_0, T)$ was calculated using the ACT/365 convention.

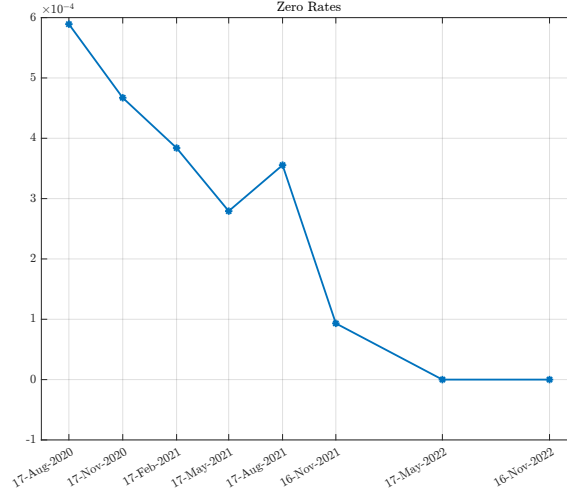


Figure 3: Left: Zero Rates.

Expiry Date T	Discount Factor $B(t_0, T)$
17 Aug 2020	0.9999
17 Nov 2020	0.9998
17 Feb 2021	0.9997
17 May 2021	0.9997
17 Aug 2021	0.9996
16 Nov 2021	0.9999
17 May 2022	1.0000
16 Nov 2022	1.0000

Table 2: Discount Factors $B(t_0, T)$ by Expiry Date

From these results, we can observe that the discount factors remain almost constant over time—a rather unusual pattern in most financial markets. After carefully analyzing this behavior, we concluded that it is likely due to the specific context of the oil market during the period under consideration. In fact, during the Covid-19 pandemic, global demand for oil dropped dramatically. Many of the typical uses of oil—most notably in transportation—were severely reduced due to lockdowns and travel restrictions. As a result, the usual dynamics that influence discounting in commodity markets were significantly disrupted. More specifically, one possible interpretation is that investors were reluctant to hold oil in their portfolios due to the high storage costs and the uncertainty about when demand, and thus prices, would start to recover.

3 Additive Bachelier Model

In this section we were asked to calibrate the Additive Bachelier model. We used the paper [13] as a guideline. Firstly we used ATM call prices, retrieved via spline interpolation on strikes and corresponding prices, for each expiry. We then used the standard bachelier pricing formula:

$$C_b(x, t; \sigma_t^b) = B_0 \mathbb{E}_0 [F_t - K]^+ = B_0 \sqrt{t} c_b \left(\frac{x}{\sqrt{t}}, \sigma_t^b \right) = B_0 \sigma_t^b \sqrt{t} c_b \left(\frac{x}{\sigma_t^b \sqrt{t}}, 1 \right)$$

$$c_b(y, \sigma) := -y \Phi \left(-\frac{y}{\sigma} \right) + \sigma \varphi \left(-\frac{y}{\sigma} \right)$$

Where:

- $x = K - F_0$: moneyness (strike minus initial forward)
- B_0 : discount factor at time 0
- σ_t^b : Bachelier volatility
- $\varphi(\cdot)$: probability density function (PDF) of the standard normal distribution
- $\Phi(\cdot)$: cumulative distribution function (CDF) of the standard normal distribution
- $c_b(y, \sigma)$: normalized Bachelier call price

We used this formula with moneyness $x = 0$, since we are taking into account the ATM calls. Equating this formula with market prices, we computed the ATM volatilities:

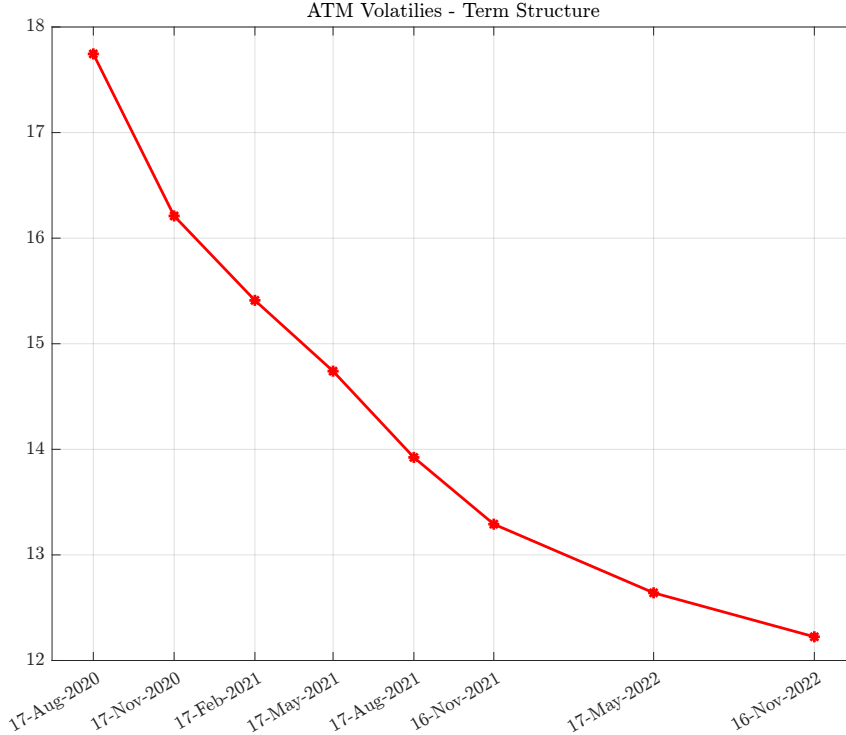


Figure 4: Zero Rates and ATM Volatility

After obtaining the ATM Bachelier volatilities, we proceeded to implement a pricing function for European call options using Lewis' formula, adapted to the *additive Bachelier framework*. The method expresses the option price in terms of the characteristic function of the underlying process, which in our case is given by:

$$\psi(u; k, \tfrac{1}{3}) = \frac{2}{k} \left\{ 1 - \left(1 + \frac{3uk}{2} \right)^{1/3} \right\}$$

Using this, the price of a call option can be computed via the following Fourier integral representation:

$$C(x, t) = B(t_0, t) \left(R_a + \frac{e^{-x}}{2\pi} \int_{-\infty}^{\infty} \phi_t(\xi - ia) \frac{e^{-i\xi x}}{(i\xi + a)^2} d\xi \right) \quad (1)$$

where:

- $x = \log(K/F_0)$: log-moneyness,
- $\phi_t(\xi)$: characteristic function of the log-price process,
- $a > 0$: damping parameter ensuring integrability,
- R_a : correction term ensuring convergence,
- $B(t_0, t)$: discount factor from t_0 to t .

To efficiently evaluate this integral, we employed the Fast Fourier Transform (FFT), a numerical algorithm that reduces the computational complexity of discrete Fourier transforms from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$.

In option pricing, the FFT is particularly useful as it allows the computation of option prices across a wide range of strikes in a single operation. By recasting the pricing problem as a convolution, it becomes FFT-compatible and thus highly efficient. In our setting, FFT enables precise and fast evaluation of option prices even in models like the additive Bachelier, which lack closed-form expressions for general strikes.

This approach offers both numerical stability and computational speed, which makes it especially valuable for large-scale pricing and calibration tasks.

Once the FFT-based pricing routine was in place, we proceeded to calibrate the model to market data. We normalized all the market prices by dividing them by $\sigma_{\text{ATM}} \sqrt{\delta(t_0, T)} B(t_0, T)$, where T denotes the expiry of the quoted options. This normalization ensures consistency across different maturities and strikes, allowing for meaningful comparison with the model outputs.

We then focused on identifying the remaining two parameters of the model, η and k . To this end, we minimized the mean squared error (MSE) between the normalized model prices—computed via FFT—and the normalized market prices. This optimization yielded the values of η and k that provided the best fit to observed data. The resulting estimates are:

Parameter	Value
κ	0.8696×10^0
η	-5.39435×10^{-3}

Table 3: Estimated parameters

We then calculated the two parameters taking into account, at each time, the call options for a certain maturity. The results we obtained are the ones described in figure (5):

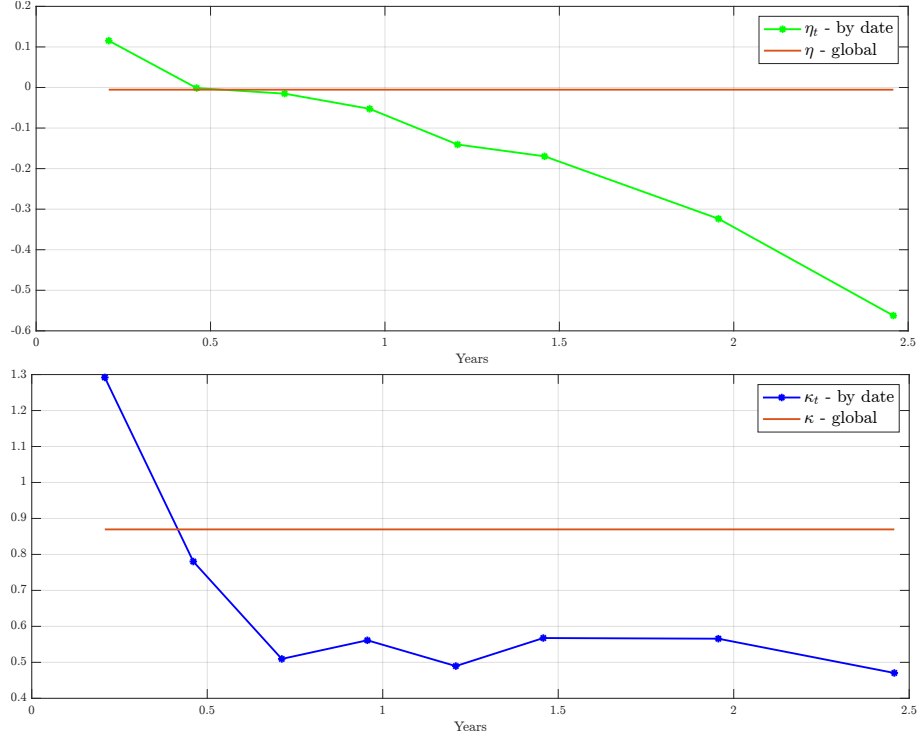


Figure 5: Top: η parameter. Bottom: κ parameter.

We then computed the MSE corresponding to each maturity, obtaining the following results:

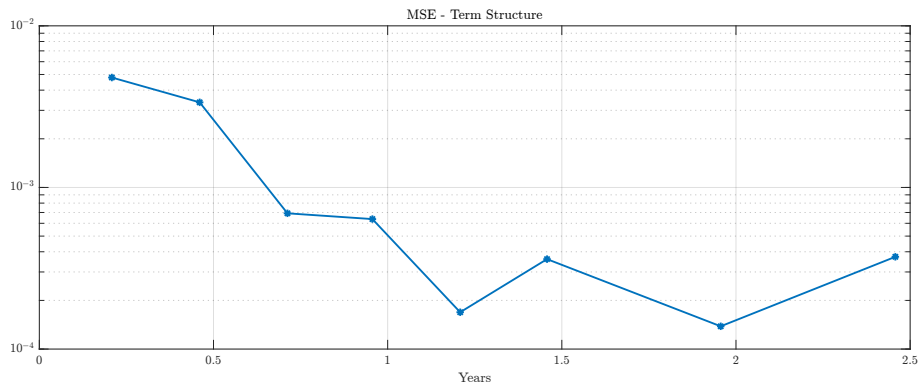


Figure 6: MSE for each Reset Date

In the final step, we computed the implied volatilities σ of the additive Bachelier model by equating the model pricing formula with the observed market prices, treating σ as the unknown variable to be solved for.

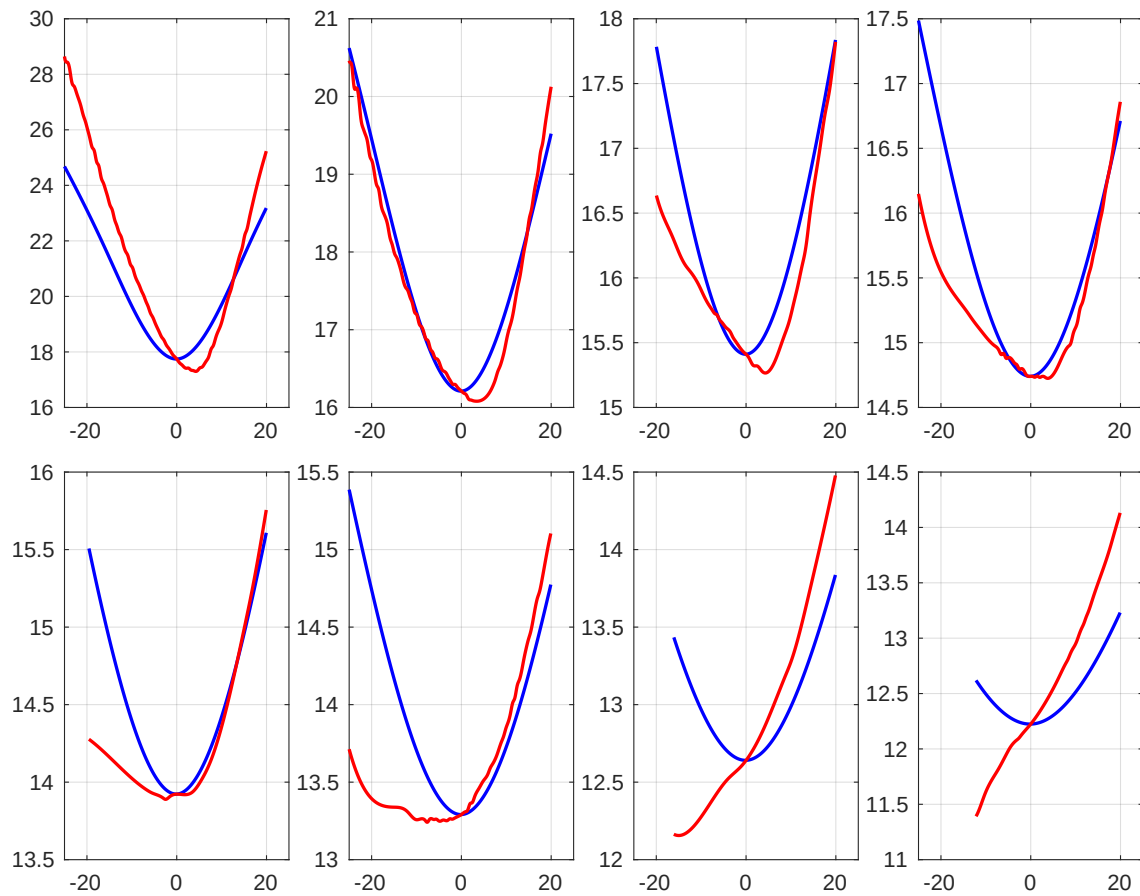


Figure 7: Additive Bachelier Implied Vol (Red) and Market Vol (blue) for each expiry. We see the Additive Bachelier is quite able to replicate the market smile.

4 Additive Logistic Model

In this section, we were asked to calibrate the Additive Logistic model, using the paper [12] as a guideline. We first considered the following pricing formula for a call option written on S^R with strike $K \in \mathbb{R}$ and maturity $T > 0$:

$$C_0^R(K, S_0^R, T) = s(T) \ln \left(1 + \exp \left(\frac{S_0^R - K}{s(T)} \right) \right), \quad S_0^R \in \mathbb{R}.$$

In particular, we adopted the functional form $s(t) = \sigma t^H$ with $\sigma > 0$ and $H > 0$, which corresponds to the *self-similar logistic additive* model (SSLA), as introduced in Corollary 4.3 of [12]. Then we calibrate the model minimizing the following expression:

$$\frac{1}{N} \sum_{i=1}^M \sum_{j=1}^{n_i} (C_{\text{mkt}}(K_{i,j}, F(t_0, T_i), T_i) - C_{\text{ssla}}(K_{i,j}, F(t_0, T_i), T_i; \sigma \cdot T_i^H))^2$$

We grouped market call prices by maturity T_i , each associated with a set of strikes $\{K_{i,j}\}_{j=1}^{n_i}$. The total number of quoted call options is $N = \sum_{i=1}^M n_i$, and we minimized the mean squared error across all prices by calibrating σ and H . The two parameters we obtained are:

H	σ
0.3349	8.3179

Table 4: Calibrated parameters of the SSLA model.

In the end, we computed the ATM implied volatility of the SSLA model as a function of time to maturity, obtaining the following term structure:

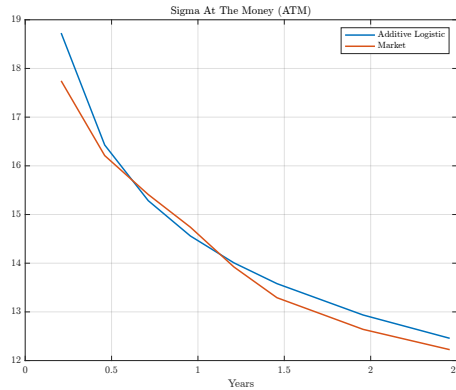


Figure 8: ATM implied volatility term structure of the SSLA model.

The model slightly overestimates short-term ATM volatilities because it imposes a smooth decay via $\sigma \cdot T^H$, while in reality, during 2020 the WTI market exhibited sudden shifts in short-term volatility, with periods of intense stress followed by rapid compressions in uncertainty. These abrupt transitions, driven by policy interventions, storage constraints, and extreme price behavior, are not captured by the model's deterministic and continuous term structure, which tends to smooth out such localized shocks.

5 A (fast) Monte Carlo simulation

The two models calibrate in the previous sections are both based on the process $\{f_s\}$ being *Additive*. For such processes, thanks to the Lévy-Khintchine formula, it's always possible to find an analytic expression for the characteristic function $\phi_t(u) = \mathbb{E}[e^{iut}]$. Moreover, the characteristic function of the increment $f_t - f_s$ between times $s < t$ is given by

$$\phi_{s,t} = \mathbb{E}[e^{iu(t-s)}] = \frac{\mathbb{E}[e^{iut}]}{\mathbb{E}[e^{ius}]}$$

We can use this relation to compute an integral expression for the Cumulative Distribution Function (CDF) of the increment $\mathbb{P}(f_t - f_s \leq x) = P(x)$, written in terms of a Fourier transform

$$P(x) = R_a - \frac{e^{-ax}}{\pi} \int_0^\infty \Re \left[\frac{e^{-iux} \phi_{s,t}(u - ia)}{iu + a} \right] du \quad (2)$$

where a is a shift factor to move the integration away from the pole $u = 0$ and R_a a correction term that depends on the shift. Given $(-p_t^-, p_t^+)$ the analyticity strip of the characteristic function, the correction term R_a is

$$R_a = \begin{cases} 1 & \text{if } 0 < a < p_t^+ + 1 \\ \frac{1}{2} & \text{if } a = 0 \\ 0 & \text{if } -p_t^- < a < 0 \end{cases}$$

in practice, we take $a > 0$. The integral above can be numerically computed by means of the Fast Fourier Transform (FFT) algorithm. Due to the high level of vectorization, this turns out to be extremely fast.

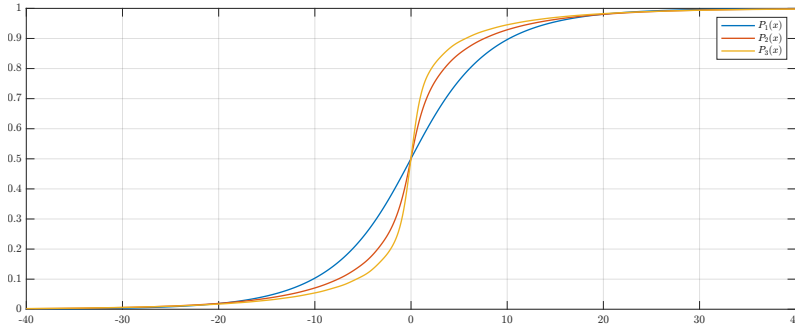


Figure 9: Numerically estimated Cumulative Distribution Function of the Additive Bachelier model, for three different points in time.

Once the cumulative distribution function (CDF) has been estimated, we follow the procedure outlined in the second appendix of [11]. It is well known that if X is a random variable with cumulative distribution function $F_X(x)$, then the transformed variable $F_X(X)$ is uniformly distributed on the interval $[0, 1]$, i.e., $F_X(X) \sim \text{Unif}([0, 1])$. Consequently, to sample from the distribution of X —which is otherwise unknown—we invert the relation by setting $X = F_X^{-1}(U)$, where $U \sim \text{Unif}([0, 1])$.

Numerically, after sampling the vector U , we interpolate it on the values of the estimated CDF $\hat{P}$, over a spatial grid $\{x_i\}$:

```
U = rand(1,N_sim);
X = spline(P,x_i,U);
```

We are now able to generate the whole path of the underlying forward, starting from an initial condition F_0 .

To check the quality of our work, we've compared the prices of European Calls, for a wide range of money-ness, obtained by the Lewis-FFT method and by the Monte Carlo pricing scheme.

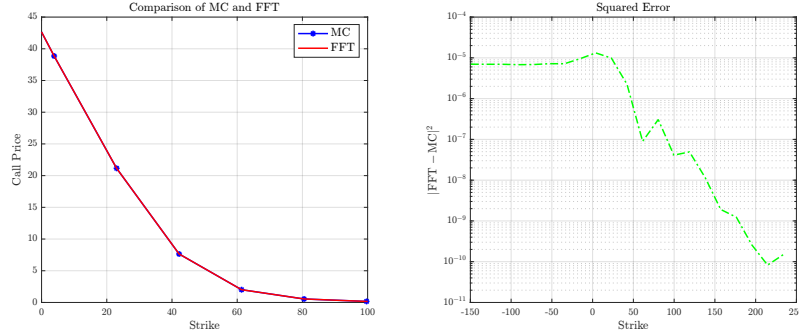


Figure 10: Left: comparison between the prices of Lewis-FFT and Monte Carlo. We see they match almost perfectly. Right: squared difference between the two prices. The error depends on the strike, but it's always low.

This simulation framework is particularly useful in pricing path dependent derivatives, like this *Asian-like* digital option. Let $F(t_i, t_N)$ be the value of the forward, expiring in t_N November 16th 2022, at time t_i . Our contract pays 1\$ if the mean of the forward is above a certain strike K , taken equal to $F(t_0, t_N)$

$$\text{Payoff} = \mathbb{I} \left\{ K > \frac{1}{N} \sum_{i=1}^N F(t_i, t_N) \right\}$$

We implemented the Monte Carlo simulation to model the evolution of the underlying asset between consecutive reset dates, in accordance with the definition of an additive process. At each step, the simulated increment was added to the value of the underlying at the previous reset date, thereby constructing the full trajectory of the asset.

After generating N_{sim} complete simulations across all reset dates, we computed the payoff of the option for each simulated path. The final option price was then estimated by averaging the payoffs over all simulations.

Model	Pice of the Option
Additive Bachelier Model	0.49925
Additive Logistic Model*	0.49720

Table 5: Prices in the two models, ADL price calculated with Python

6 Hedging an Asian Option

Once the exotic option was priced, it became critical to implement a hedging strategy aimed at lowering the sensitivity of the portfolio to changes in model parameters volatility σ , vol of vol κ , skewness η and the underlying asset(F).

For this purpose, we constructed a specific hedge using plain vanilla options along with forward contracts, all set to the same time to maturity as the exotic option. The Greeks were computed numerically using finite derivative methods.

The remaining linear system could be solved to find the optimal amounts of vanilla options and forwards to construct an effective hedge, since ATM options are not traded in the market we select the nearest strike from the ATM strike, so we select Put options and Call Options with strike 40 and 42.

The strategy described above, as depicted in the graph below, give us a successfully performed hedging strategy: significantly curtailing portfolio return volatility amply and maintaining the portfolio at a reasonable surcharge throughout the 10-day period with an eventual net profit.

This also brings to the final point, employing static over dynamic hedging strategies, which offered better long-term solutions considering the fixed costs associated to trading options frequently coupled with market instability.

Type	Quantity	Operation
Forward	218,043	Long
Call Options	-1,153,265	Short
Put Options	1,373,965	Long
Fee cost of the Strategy		11,659.98

Our hedging strategy achieved a total return of 2.74% over the course of the following two weeks.

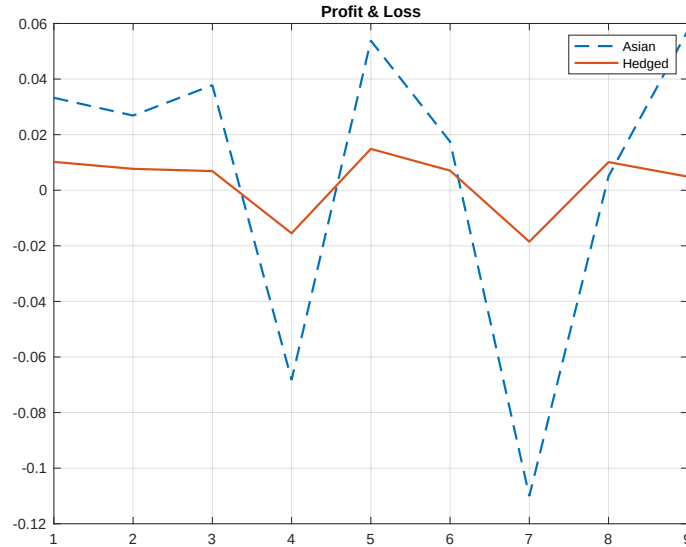


Figure 11: Daily returns of the exotic product (blue, dotted line) and the hedged portfolio (red, continuous line). The hedging strategy is much less volatile than the uncovered position.

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